

Course Code: CSE255	Course Title: ANALYSIS OF ALGORITHMS LAB Type of Course: Practical	L- C	T- 0	P- 0	2	1
Version No.	2.0					
Course Pre-requisites	Meaning of Analysis and various analysis and its extension, Mathematical Induction and its importance to analysis of Algorithms, Introduction to Pseudo code, Knowledge of Recursive and Non Recursive algorithms.					
Anti-requisites						
Course Description	This Course introduces techniques for the design and analysis of efficient algorithms and methods of applications. It deals with analyzing time and space complexity of algorithms, and to evaluate trade-offs between different algorithms. Topics include: Brute force- Bubble sort, linear search, Divide-and-conquer- Merge sort, Quick sort. Dynamic programming and greedy technique- Prim's, Kruskal's, Dijkstra's Algorithm, Warshall's algorithm, Floyd's algorithm, Coin changing problem, Multi stage graph – Optimal Binary Search Trees ,Backtracking – N Queens Problem, Hamiltonian Path Problem, M Coloring Problem. Backtracking.					
Course Objective	The objective of the course is to familiarize the learners with the concepts of Analysis of Algorithms Lab and attain SKILL DEVELOPMENT through EXPERIENTIAL LEARNING techniques.					
Course Out Comes	On successful completion of the course the students shall be able to: Compute time complexities for various Recursive and non-recursive Algorithms [Application]. Demonstrate the Brute Force Technique for real world problems [Application] Apply divide and conquer technique for searching and sorting [Application] Demonstrate the Dynamic Programming and Greedy Algorithms for various applications [Application]					
Course Content:	Non-recursive algorithms: Factorial, Max. Recursive algorithms: Factorial, GCD, Search, Tower of Hanoi. Brute Force Technique: Bubble sort, Linear Search. Divide and Conquer: merge sort, quick sort. Dynamic programming: Coin changing problem, Multi stage graph – Optimal Binary Search Trees ,The knapsack problem, Warshall's Algorithm, Floyd's Algorithm. The Greedy Method: Prim's and Kruskal's algorithm to find Minimum Spanning Tree, Single source shortest path (Dijkstra's Algorithm), Boolean Satisfiability Problem (SAT). Hamiltonian Path Problem, M Coloring Problem. Backtracking: N-Queens problem.					

List of Laboratory Tasks

Apply non recursive algorithmic designing technique to solve Factorial of a number, Linear Search , finding max element problem and calculate the time efficiency (best, average & worst).

Apply recursive algorithmic designing technique to solve Factorial, GCD, , Tower of Hanoi, problems and calculate time (Best, average & worst) efficiency.

Apply Brute force algorithmic designing technique to sort elements using bubble sort algorithm and calculate time (Best, average & worst) efficiency.

Apply divide and conquer algorithmic designing technique to sort elements using merge sort algorithm and calculate time (Best, average & worst) efficiency.

Apply divide and conquer algorithmic designing technique to sort elements using Quick sort algorithm and calculate time (Best, average & worst) efficiency

Apply dynamic programming algorithmic designing technique to find All pair Shortest Path for a given graph using Floyds and Warshall's algorithm

Apply dynamic programming algorithmic designing technique for Solving 0/1 knapsack problem and find its efficiency.

Apply dynamic programming algorithmic designing technique for Solving Coin changing problem and find its efficiency.

Apply dynamic programming algorithmic designing technique to find Optimal Binary Search Trees.

Apply greedy algorithmic designing technique for constructing MST for a given graph using prim's algorithm

Apply greedy algorithmic designing technique for constructing minimum spanning tree using Kruskal's algorithm

Apply backtracking algorithmic designing technique for M Coloring Problem

Apply backtracking algorithmic designing technique for solving queen's problems for 4, 8 and 16 inputs.

Targeted Application & Tools that can be used:

Social media networks, GPS applications, Google search, e-commerce platforms, Netflix recommendation systems, etc.

Text Book

Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, "Introduction to Algorithms", PHI Learning Private Limited.

References

Anany Levitin, "Introduction to the Design and Analysis of Algorithms", Pearson Education, 3rd edition.

Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, "Data Structures and Algorithms", Pearson

E-Resources

NPTEL course – <https://nptel.ac.in/courses/106106131>

Topics relevant to the development of SKILLS:

Quick sort

The knapsack problem

Warshall's Algorithm

Floyd's Algorithm.

Prim's and Kruskal's algorithm to find Minimum Spanning Tree

Single source shortest path (Dijkstra's Algorithm).

Backtracking: N-Queens problem.

for Skill Development through Experiential Learning techniques. This is attained through assessment component mentioned in course handout.